University of Cape Town



Prescriptive Analytics Assignment

Soft Systems Methodology and Cognitive Mapping

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The Messy Problem

Competing Interests

A telecommunications company must decide which network sites to upgrade over the next five years in order to remain competitive in a fast-changing and highly contested market. These decisions must balance the coverage needs of customers across different socio-economic strata, including a commitment to maintaining quality coverage in rural and remote communities, even where these are not the most revenue-generating. They are further constrained by legal and regulatory conditions governing the spectrum available to the company, intended to ensure fair competition, and by how that spectrum is distributed across sites already in the network.

At the same time, the company has set a strategic ambition to transform from a traditional telecoms operator into a leader at the forefront of industry innovation, in line with increasing shareholder value. This has driven a growing push to adopt AI across its operations and business strategy, requiring significant capital expenditure that competes with network investment. The push has also generated unease within the workforce, where many employees fear that AI will make their roles redundant and feel they have not been brought along or shown a place in the company's next strategic direction. Additionally, the company's leadership culture relies heavily on positional power and top-down decision-making, which slows feedback loops and limits the extent to which employee concerns and frontline knowledge can inform strategic choices.

Key Conflicts:

- Competitive Edge vs Social Responsibility
- Technology Adoption vs Employee buy-in
- Regulatory requirements vs Shareholder Value (Continued Revenue Growth)

Problem Expression

System Root Definition

A company-owned system, envisioned executives, that prioritizes technology expansion and AI adoption to grow company value, ensure that the company remains competitive and viable over the next five years inline with fiduciary obligations to shareholders.*comeback

Customers: Subscribers, Shareholders

Actors: Network Planners, Engineers, and C-Suite Executives

Transformation: Improved network service through the upgrade of existing sites and expanded coverage for remote areas

Worldview (from executives' perspectives): quality coverage and innovation are both necessary to remain viable

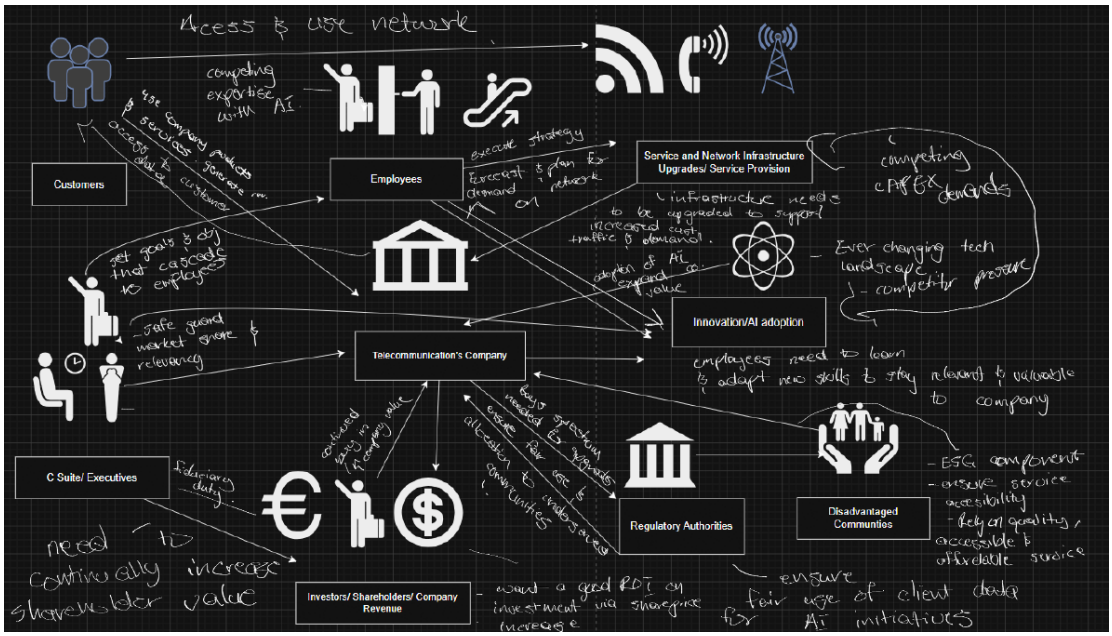
Owners: Company C Suite, set the business strategy and cascade goals to different parts of the company. Decide on the allocation of strategic investments and expenditure that the company adopts. Have a fiduciary obligation to shareholders.

Environmental constraints: spectrum regulation, capital limits, competitors, technology change

Subsystem Root Definition (Network Engineers and Planners) * comeback

A technical system owned by by network planners, engineers that priorities and upgrades network sites to ensure growing demand is met and deliver excellent service to network customers. within regulatory spectrum limits, in order to transform an uneven network and an uncertain strategic direction into a network that delivers quality service to urban and rural subscribers alike,

Rich Picture the problem



Activities Required to make the Root Definition True

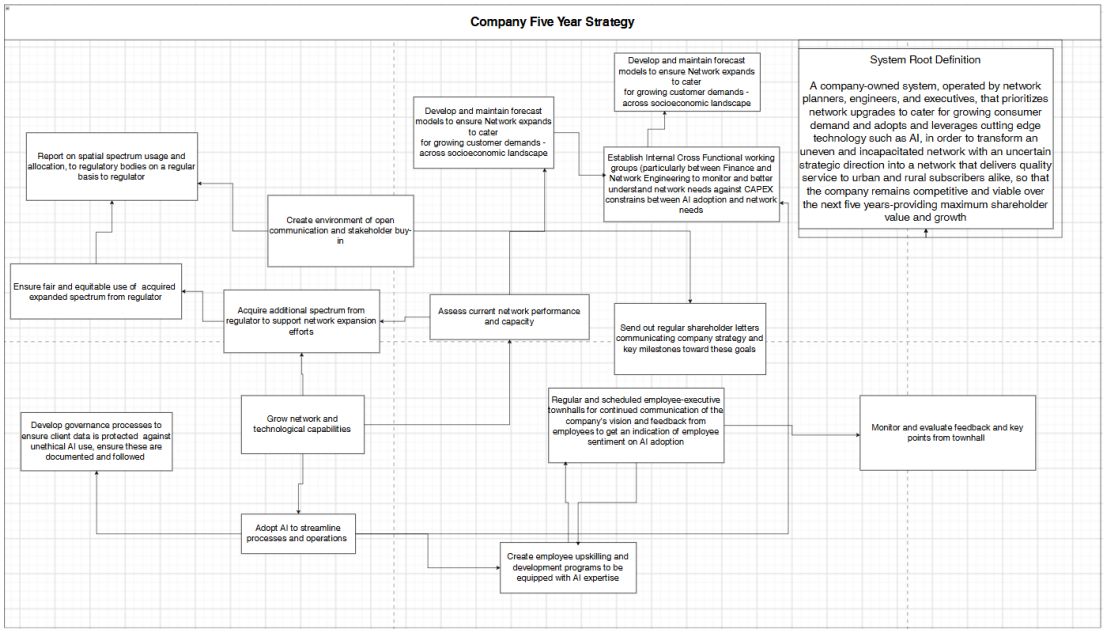
- Company initiatives and employee development program to use and leverage AI in daily working processes and operations. Posit the use of AI and an enhancer of the work process for employees, and not a replacement.
- Regulatory consultations and reporting to ensure network technology expansion is in line with legal requirements as well as ESG obligations w.r.t rural network accessibility.
- Use cross-functional working- specifically groups to equip and educate network planners and engineers to understand and optimize for financial constraints that required network upgrades are under and educate the finance department of the needs to ensure that the network can support increasing customer demand
- Continuous engagement between employees and executives via town halls and workshops to secure employee buy-in in the broader vision and goals of the company and to ensure that executives are in the loop w.r.t employee sentiment (creates environment of open communication and feedback).

Combined Root Definition:

System Root Definition

A company-owned system, operated by network planners, engineers, and executives, that prioritizes network upgrades to cater for growing consumer demand and adopts and leverages cutting edge technology such as AI, in order to transform an uneven and incapacited network with an uncertain strategic direction into a network that delivers quality service to urban and rural subscribers alike, so that the company remains competitive and viable over the next five years-providing maximum shareholder value and growth.

SSM Model


Company Five Year Strategy
System Root Definition
A company-owned system, operated by network planners, engineers, and executives, that prioritizes network upgrades to cater for growing consumer demand and adopts and leverages cutting edge technology such as AI, in order to transform an uneven and incapacitated network with an uncertain strategic direction into a network that delivers quality service to urban and rural subscribers alike, so that the company remains competitive and viable over the next five years-providing maximum shareholder value and growth
Develop and maintain forecast models to ensure Network expands to cater for growing customer demands - across socioeconomic landscape
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Establish Internal Cross Functional working groups (particularly between Finance and Network Engineering to monitor and better understand network needs against CAPEX constrains between AI adoption and network needs
Report on spatial spectrum usage and allocation, to regulatory bodies on a regular basis to regulator
Create environment of open communication and stakeholder buy-in
Ensure fair and equitable use of acquired expanded spectrum from regulator
Acquire additional spectrum from regulator to support network expansion efforts
Assess current network performance and capacity
Send out regular shareholder letters communicating company strategy and key milestones toward these goals
Develop governance processes to ensure client data is protected against unethical AI use, ensure these are documented and followed
Grow network and technological capabilities
Regular and scheduled employee-executive townhalls for continued communication of the company's vision and feedback from employees to get an indication of employee sentiment on AI adoption
Monitor and evaluate feedback and key points from townhall
Adopt AI to streamline processes and operations
Create employee upskilling and development programs to be equipped with AI expertise